

Standardizing Hybrid Automata: From
Theoretical Foundations to Real-World
Implementation Frameworks

A Pilot Literature on the Development of a Unified Framework for
Hybrid Automata Development and Simulation

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Abstract

Hybrid automata are formal models for a dynamical system with discrete and continuous components [1] widely adopted across research & industry for a variety of use cases ranging from control systems to stock trading algorithms, and from robotics to cyber-physical systems a recent example of this is for solving COLREG compliant autonomy for maritime vehicles [2] where the hybrid automata is used to model the discrete decision making process of the vessel and the continuous dynamics of the vessel's movement.

This theoretical framework provides a powerful method for modeling complex systems that can solve real world problems algorithmically without the need for human intervention during the execution of the system in a way that is efficient and effective and totally transparent for the human-in-the-loop to comprehend the decision making process of the system.

While in recent years there has been significant progress in the development of Artificial Intelligence (AI) through the rise of large-scale models like OpenAI's GPT series [?] that have shown the ability to imitate human-like reasoning further blurring the lines between human and machine intelligence therefore reducing the gap between what humans and machines can do there is still issues with AI that make it difficult to deploy in real world application such as lack of transparency and explainability due to the black-box nature of decision making caused by the topology of deep-neural networks (dnn) and also an issue with the reliability of AI systems due to their sensitivity to mistakes in novel situations that they have not been trained on and undefined emergent behaviours arising from outliers in their training data.

for the reasons presented above, hybrid automaton is still one of the most powerful tools for complex system design however, the theoretical framework has been laid there seems to be a lack of standardization in converting the theoretical framework into real world applications with several papers implementing their own versions of hybrid automaton with different algorithms enabling them.

In this pilot literature review, we will explore the current state of the art in hybrid automata development and simulation to identify the current gaps in the field and propose a unified framework for hybrid automata development and simulation that can be adopted across research and industry to standardize the development process and enable more efficient and effective use of hybrid automata in real-world applications.

0.1 introduction

something happens in here

Bibliography

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